

Algebra 1
Unit 13
Lesson 1 Practice

Name Answer Key
Date _____
Period _____

The Yang family is adding a second floor to their home. They were approved by their bank for a loan of \$100,000 with a simple interest rate of 4.85%.

1. Write an equation that models the Yang's loan.

$$A = 100,000(1 + .0485t)$$

2. Rewrite the equation in slope-intercept form.

$$A = 4850t + 100,000$$

3. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept	100,000	Initial loan amount
slope	4,850	Yearly interest

4. What is the total the Yang's will have to pay if they pay off the loan in 10 years?

$$A = 4,850(10) + 100,000$$

$$48,500 + 100,000$$

$$\boxed{\$148,500}$$

5. How many years did it take the Yang's to pay off the loan if the total they paid was \$135,000?

$$\begin{array}{r} 135,000 = 4,850t + 100,000 \\ -100,000 \quad -100,000 \end{array}$$

$$\frac{35,000}{4850} = \frac{4850t}{4850}$$

$$7.2 \approx t$$

$$\boxed{\text{About 7.2 years}}$$

Mekhi just graduated college and is starting his first job. He is investing money into a savings account to plan for retirement. He has \$15,000 to invest in a savings account with a 3.25% APR.

6. Write an equation that models Mekhi's savings account.

$$y = 15,000(1 + .0325t)$$

7. Write the equation in slope-intercept form.

$$y = 487.5t + 15,000$$

8. Complete the table.

Key Feature	Value	Interpretation in Context
y-intercept	15,000	Initial investment
slope	487.50	Interest per year

9. How much money will be in the savings account if Mekhi wants to retire 30 years later?

$$15,000 + 487.5(30) = y$$

$$15,000 + 14,625 = y$$

$$\$29,625 = y$$

10. How many years would Mekhi have to work to retire with \$35,000?

$$35,000 = 15,000 + 487.5t$$

$$20,000 = 487.5t$$

$$41 \approx t$$

years